

IO SKY

NOTIFICATION CENTER

Functional Specification

Developer Handoff

Role-aware · Permission-aware · Scope-aware · Lifecycle-aware

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1. Purpose

The IO SKY Notification Center is the central attention and action layer for platform users.

Its purpose is not to display every event occurring inside the platform. Its purpose is to show each authorised user only the events that are relevant to that user's responsibilities, permissions, assigned resources and required actions.

The Notification Center must therefore be role-aware, permission-aware, scope-aware, assignment-aware, ownership-aware, lifecycle-aware, resource-aware and action-oriented.

The Notification Center must remain compact enough for quick operational use. Detailed information belongs inside the underlying authorised record rather than inside the notification itself.

2. Notification Center vs Audit Log

The Notification Center and Audit Log are separate platform concepts and must not be merged.

Notification Center - purpose: user awareness, preparation and action. Examples include a Discovery Call being scheduled, an AI Scan being ready for review, a project milestone requiring attention, a Client supplying requested information, or a security event requiring authorised action.

Audit Log - purpose: traceability, security, governance and historical accountability. Examples include who changed a permission, when a record was modified, who published a Scan, when privileged access was granted, or when configuration changed.

A system event may produce both an audit record and a notification, but this does not make them the same object. Routine audit activity must not automatically generate Notification Center noise.

3. Notification Access Principle

A notification must never grant access to the underlying resource. The Notification Center only exposes information that the recipient is already authorised to see. When a user opens a notification, backend authorisation must be checked again.

Access must therefore be calculated from: Role + Permission + Scope + Assignment/Ownership + Resource + Lifecycle State. Where relevant, environment must also be part of the authorisation calculation.

This protects against situations where a user received a notification before access was revoked, a Client or Project assignment changed, the underlying resource moved into a different lifecycle state, permissions changed, or a user attempts direct URL access. Frontend visibility alone is never sufficient authorisation.

4. Notification Bell

The platform header contains a notification bell. The bell acts as the entry point to the Notification Center and must provide an immediate indication when relevant unread or attention-required notifications exist.

The bell must not become a counter for every background event. Its purpose is to signal that there is something relevant for this user to review, prepare for or act on.

5. Notification Dropdown

Opening the bell displays a compact dropdown intended for quick situational awareness and preparation. It must not attempt to display the complete underlying record.

Each notification row should provide only enough information for the recipient to understand what happened, what resource it relates to, when it happened or requires attention, whether action is required, its current relevant state, and where the user can go next. Selecting the row takes the user to the authorised destination.

6. Notification Row Structure

Each notification row must support the following logical fields where applicable:

- Event type / title: a concise description such as New Discovery Call, AI Scan Ready for Review, Project Milestone Updated, or Client Information Required.
- Context: relevant Client, prospect, Project, AI Scan, invoice/payment, system/integration, environment or other authorised resource.
- Date / time: scheduled time, creation time, review deadline, milestone date or payment due date.
- Lifecycle state: for example Scheduled, Submitted, Review Required, Awaiting Approval, Published, Action Required, Resolved or Cancelled.
- Attention / priority: where appropriate, reflecting genuine operational significance rather than decoration.
- Authorised action: for example View Call, Review Scan, Open Project, View Client, Review Billing or Open Security Event.

The notification itself does not perform privileged actions without normal authorisation controls.

7. Read State and Action State

Read/unread status and operational completion are different concepts. A notification can be unread but already resolved, or read but still requiring action.

Opening a notification therefore does not automatically mark the underlying operational task as completed. The platform must distinguish between read state and attention/action state.

8. Resolved, Cancelled and Superseded Events

Notifications must remain lifecycle-aware. If the underlying event has been resolved, cancelled, superseded, completed or otherwise rendered no longer actionable, the Notification Center must not continue presenting it as unresolved attention.

Historic information can remain available where appropriate, but its state must accurately reflect reality.

9. Notification Families

The Notification Center must support structured Notification Families used for routing and filtering:

1. Discovery Calls
2. AI Scans

- 3. Clients & Accounts
- 4. Projects & Delivery
- 5. Payments & Billing
- 6. AI Operations
- 7. Platform & Integrations
- 8. Security & Access
- 9. Governance / System

These families are event classifications, not user roles.

10. Discovery Calls Notification Family

This family covers relevant Discovery Call lifecycle events, including new requests, scheduled calls, rescheduled calls, cancellations, same-day retry/follow-up where applicable, operationally relevant prospect updates and lifecycle changes requiring attention.

Where permitted, a row may include prospect/client name, company, scheduled date/time, first-call or follow-up/retry context and current call state. Complete prospect and intake information remains inside the Discovery Call record.

11. AI Scan Notification Family

This family covers the operational and review lifecycle of Operations Scan, Cyber Scan and Elite Scan.

Possible events include Scan purchased/created, intake started where operationally relevant, required Client information missing, Scan submitted, analysis completed, Scan ready for human review, review required, reviewer assigned, review completed, approval required, publication required, Scan published, Client-facing Scan available and Client action/information required.

Not every Scan event needs to generate a notification for every user. Routing depends on responsibility.

12. AI Scan Human Review Notification Rules

Generated Scan output must never automatically become visible to the Client.

Required lifecycle: Assessment → Analysis → Human Review → Approval → Publish → Client Portal.

When analysis is ready, the relevant authorised reviewer may receive AI Scan Ready for Review. After review, the authorised publishing user may receive AI Scan Ready for Publication. A Client notification may only be generated after the Scan has reached its authorised published state.

Internal review findings and draft output must never be exposed through Client notifications.

13. Clients & Accounts Notification Family

This family covers relevant authorised Client/account lifecycle events, including Client account creation, operationally relevant Client information updates, Client action required, Client submission of required information and account-related events requiring authorised attention.

Notifications must respect Client assignments. An Admin managing Client A must not receive Client B notifications simply because both exist in the platform.

14. Projects & Delivery Notification Family

This family covers relevant Project and delivery events, including assignment/reassignment, milestone changes, approaching milestones, delivery-state changes, review requests, reviewer assignments, dependencies, blockers, blocker resolution, deadlines, Client information requests, relevant project communication and authorised delivery-state changes.

Notifications must follow Project assignment and permissions.

15. Payments & Billing Notification Family

This family covers authorised billing-related events such as invoice creation where attention is required, invoice/payment status changes, operationally meaningful payment completion, billing action required, payment issues and authorised account billing events.

Billing notifications are visible only to users with appropriate Billing permissions. Billing access is not automatically inherited from general Admin access.

IO SKY does not require a refund or chargeback workflow or Notification Family. Such functionality must not be introduced unless separately instructed in the future.

16. AI Operations Notification Family

This family covers operational AI-system events that genuinely require human awareness or intervention, such as an AI process unable to continue, required human intervention, a processing exception, an approved operational threshold being exceeded, or AI service unavailability with operational impact.

Routine successful AI processing should not generate notifications. The Notification Center must not become a log of every AI action.

17. Platform & Integrations Notification Family

This family covers relevant system and integration events requiring attention, such as integration disconnection, expired authentication, integration failure affecting operation, external service unavailability, platform components requiring intervention or configuration issues affecting functionality.

Successful syncs, successful scheduled jobs and ordinary healthy background activity should not generate unnecessary notifications.

18. Security & Access Notification Family

This family contains sensitive security and access events, such as relevant security alerts, privileged access granted/revoked, significant permission changes, important access events, Security Center action required and security configuration requiring authorised attention.

Visibility must be tightly controlled. Technical Operator status does not automatically grant access to this family. Security-related notifications require corresponding authorised Security scope/permission. General Admin status does not automatically provide Security Center visibility.

19. Governance / System Notification Family

This family is primarily relevant to Super Admin oversight and other specifically authorised governance responsibilities. Events may include significant privileged-access changes, important governance events, administrative configuration requiring oversight, high-level platform issues and relevant governance action required.

Routine configuration activity belongs primarily in the Audit Log and should only produce a Notification Center event when human attention is genuinely required.

20. Super Admin Notifications

Super Admin receives the broadest relevant notification coverage across Discovery Calls, AI Scans, Clients & Accounts, Projects & Delivery, Payments & Billing, AI Operations, Platform & Integrations, Security & Access and Governance/System.

Broad visibility does not mean every successful background action should generate a notification. Anti-noise principles continue to apply.

21. Admin Notifications

Admin status alone is not sufficient to determine notification visibility. Effective notification routing must follow the Admin's permissions, assigned preset, subsequent permission customisations, Client assignments, Project assignments, resource scopes and operational responsibilities.

For example, a Discovery Call Manager may receive Discovery Call notifications but no unrelated Billing, Project, Scan Review or Security notifications. An AI Scan Reviewer may receive review notifications for assigned Scans without general operational Admin notifications. A Billing Manager may receive billing notifications without internal Scan findings.

The preset provides a starting configuration only. Final routing follows effective permissions and assignments.

22. Developer Notifications

Developers receive only notifications connected to authorised development responsibilities, such as Project/work assignment, implementation assignment, reviewer assignment, review requested, deadline, dependency, blocker, relevant deployment/review events and authorised Project communication requiring Developer attention.

Developers do not automatically receive Discovery Call, general Client, sales/prospect, billing, unrelated Scan-review, general Security Center or governance notifications.

23. Implementer and Reviewer Notifications

Developer responsibilities distinguish between Implementer and Reviewer and must influence notification routing.

The Implementer may receive implementation assigned, requirements updated, blocker/dependency, reviewer feedback and implementation action required.

The Reviewer may receive review requested, implementation ready for review, review deadline and required review action.

A person should not receive Reviewer notifications solely because they are a Developer. The actual assignment must exist.

24. Technical Operator Notifications

Technical Operator notification visibility must follow assigned technical scopes: Frontend, Backend, Full Stack, UI/UX, Infrastructure / DevOps, Security and Database / Data.

A Frontend Technical Operator may receive relevant frontend events for an assigned Project but does not automatically receive database alerts, Infrastructure/DevOps notifications, Security Center notifications or unrelated Backend events.

Technical scope, Project assignment, environment and permission must all be considered where applicable.

25. Client Notifications

A Client receives only notifications connected to their own authorised Client account.

Possible Client-facing events include approved AI Scan available, Project update, milestone update, information/document request, relevant communication, meeting/scheduling event, relevant billing/account event, Client-facing account/security event and action required from the Client.

Clients must never receive internal AI Scan analysis/review, unpublished findings, internal recommendations, Developer work, Technical Operator activity, internal governance, internal Security Center events or other Clients' information.

26. Permission Changes and Existing Notifications

Notification access must be evaluated dynamically where appropriate. If a user loses access to an underlying resource after receiving a notification, that notification must not allow the user to regain or bypass access.

Attempting to open it must trigger normal backend authorisation. The development team may determine the appropriate handling of inaccessible stale notifications provided no data exposure occurs.

27. Assignment Changes

Notification routing must respond correctly when responsibility changes. Future notifications must follow the current Scan Reviewer, Developer Reviewer, Client assignment, Project assignment or other relevant responsibility rather than relying on an old recipient list.

28. Deep Links

Where applicable, a notification row should direct the user to the relevant authorised location in the platform: Discovery Call record, internal Scan review workspace, Client-facing published Scan/report, Project/milestone, authorised billing record or authorised security record.

Every destination remains subject to backend authorisation.

29. Notification Content and Sensitive Data

Notification previews must use data minimisation. Rows should show enough information to identify the event without unnecessarily exposing sensitive information.

Sensitive details should remain inside the underlying record. This is particularly important for security findings, AI Scan findings, technical credentials, billing information and sensitive Client information.

Credentials, passwords, secrets, tokens or equivalent privileged values must never appear in notification content.

30. Notification Noise Control

Notifications should generally exist because the user must act, should prepare, something materially changed, responsibility was assigned, an exception occurred, a significant lifecycle state changed, a deadline/dependency requires attention, or there is a meaningful security/operational issue.

Routine healthy operation generally belongs elsewhere. Successful background syncs, normal autosave, successful scheduled jobs, normal AI processing steps, ordinary database operations, every minor field update and every audit event should normally not generate notifications.

31. Notification Priority

Where priority is supported, it should represent operational significance rather than cosmetic styling. Priority logic should remain aligned with genuine event significance and should not be used decoratively.

32. User Notification Preferences

User preferences may control optional delivery channels or non-critical notification presentation where appropriate. They must not override mandatory security/governance requirements or change the user's permissions. Notification preference and authorisation are separate concepts.

33. Notification Delivery Channels

The Notification Center is the in-platform notification layer. Where email or another authorised channel is also used, it should follow the same permission/security principles and data-minimisation rules, routing the authorised user back into the secure platform where appropriate.

34. Notification Lifecycle Model

A notification should support a read/delivery lifecycle separately from the underlying operational attention lifecycle. Exact database implementation remains the development team's responsibility. The functional requirement is that read state and operational resolution remain distinguishable.

35. Event Deduplication

The Notification Center should avoid duplicate notifications for the same underlying event where duplication adds no value. Repeated retries of one background process should not necessarily create multiple identical rows. Genuinely new actionable information must not be suppressed.

36. Notification Ordering

Default ordering should prioritise relevance and recency while allowing genuinely urgent actionable items to remain visible. Low-value informational events should not push important unresolved attention out of practical view.

37. Security Enforcement

Notification filtering must occur on the server/backend-authorisation layer. Frontend filtering alone is not acceptable.

The system must prevent cross-Client leakage, cross-Project leakage, unauthorised Scan visibility, unauthorised Billing visibility, unauthorised Security visibility, Technical Operator cross-scope leakage and stale notification links bypassing revoked permissions.

38. Admin Permission Presets and Notifications

The ten Admin Permission Presets must integrate with Notification Center routing:

1. Client Manager
2. Project Manager
3. Discovery Call Manager
4. AI Scan Manager
5. AI Scan Reviewer
6. Billing Manager
7. Communications Manager
8. Delivery Manager
9. Operations Manager
10. Administrative Manager

Routing must follow effective final permissions, not merely the preset name. Removing Billing must stop Billing notifications. Adding AI Scan Review may enable relevant review notifications only for resources to which the Admin is assigned.

39. Scan Publish Security

Scan Publish requires special treatment. A reviewer being able to review a Scan does not automatically mean that reviewer may publish it.

scan.review and scan.publish must be independently controllable permissions.

A Scan Ready for Publication notification may only route to users with the required permission and access to that specific Scan. A Client notification must not be created before successful authorised publication.

40. Security Center Notifications

Security Center visibility must not be inferred merely from Admin role, Technical Operator role, Operations responsibilities or Project access.

Security Center access requires explicit authorised Security permissions/scope. The same applies to Security & Access notifications that expose restricted security information.

41. Environment-Aware Technical Notifications

Where technical events concern a specific environment, routing must respect environment access, including Development, Staging and Production.

A Technical Operator authorised only for Staging must not receive or gain access to Production information simply because they receive other technical notifications. Production-sensitive notifications require necessary Production access.

42. Notification Center UX Principle

The Notification Center should feel like an operational control surface, not a social-media feed.

Its design should prioritise clarity, calmness, quick recognition, preparation, action and low noise. Functional clarity is more important than decorative complexity.

43. Empty / No-Attention State

When there is nothing requiring attention, the Notification Center should communicate this clearly. It must not manufacture activity or retain obsolete unresolved indicators merely to avoid appearing empty.

An empty Notification Center can correctly mean: There is currently nothing requiring your attention.

44. Core Acceptance Rules

The Notification Center implementation is not complete unless:

1. Users only receive notifications they are authorised to know about.
2. Notifications do not grant access.
3. Permissions are revalidated when the underlying resource is opened.
4. Routing follows effective permissions, scopes and assignments.
5. Admin presets do not operate as hard-coded security roles.

6. Technical Operator technical scopes affect routing.
7. Security Center notifications require separate Security authorisation.
8. Client notifications never expose internal information.
9. AI Scan output remains internal until human review and authorised publication.
10. Scan Review and Scan Publish are independently controllable.
11. Read and resolved states are distinguishable.
12. Cancelled/resolved/superseded events do not remain falsely actionable.
13. Routine successful background events do not create excessive noise.
14. Notification Center and Audit Log remain separate.
15. Cross-Client, cross-Project and cross-environment information leakage is prevented.
16. Permission revocation prevents subsequent resource access even when an older notification exists.
17. Notification content follows data-minimisation principles.
18. The system supports the defined Notification Families.
19. Routing works across Super Admin, Admin, Developer, Technical Operator and Client.
20. The architecture remains extensible for future authorised event types.

45. Final Functional Rule

Show the right person the right event at the right time - only when they are authorised to know about it and when the event is genuinely relevant to their responsibility.

The Notification Center must help users understand what requires attention. It must never become an alternative authorisation system, an unrestricted activity feed or a substitute for the Audit Log.